THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NJOMBE TOWN COUNCIL

FORM FOUR PRE-MOCK EXAMINATION APRIL -2024

033 BIOLOGY 1

TIME: 3:00 Hours **17th April 2024**

INSTRUCTIONS

1. This paper consists of sections A, B and C with a total of ELEVEN (11) questions.
2. Answer all questions in section A and B and only two questions in section C.
3. Section A carries Sixteen (16) marks while section B carries Fifty-four (54) marks and section C carries Thirty (30) marks
4. All writing must be in blue or black ink except drawings which must be in pencil
5. Write your index number on every page of your answer booklet(s)

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SECTION A (16MARKS)

Answer all questions from this section

1. For each of the items (i) – (x) choose the correct answer among the given alternatives and write its letter beside the item number in your answer sheet provided.

i. In animal cell the nucleus perform a lot of activities which ensure life process, which one among of the following is not the function of the nucleus?
   A. Determine the chemical process that takes place in a cell.
   B. Initiate cell division.
   C. Control function of all parts of the cell.
   D. Determine the heredity characteristics of the cell.
   E. Forms cell membrane.

ii. With the regard to theories of evolution; which of the following does not show homology?
   A. Limb structures in vertebrates
   B. Beaks in birds
   C. Foot structure in birds
   D. Wings of birds and insects
   E. Pentadactyl structures in vertebrates.

iii. Epiphytes are organisms that grow on tree branches and normally do not harm the tree. Therefore, these organisms undergo which mode of feeding;
   A. Holozoic
   B. Parasitism
   C. Saprophytism
   D. Mutualism
   E. Commensalism.

iv. Despite of both pigeon and bat having wings and ability to fly, they are placed in different classes during classification. In which way pigeon can be distinguished from bat?
   A. Pigeon has fur while bat has feathers
   B. Pigeon lay eggs while bat give birth
   C. Pigeon is cold blooded while bat is warm blooded
   D. Pigeon has claws while bat has no claws
   E. Pigeon has simple eyes while bat has compound eyes

v. During respiration which substance is produced in animals when food substance especially glucose is broken down in absence of oxygen?
   A. Alcohol
   B. Lactic acid
   C. Carbon monoxide
   D. Oxygen
   E. Ethanol

vi. In hormonal coordination glands are considered to be the effector and help the body to take action. Which of the following conditions would cause the adrenal gland to produce adrenaline hormone
   A. Eating tomato
   B. Seeing a snake
   C. Smelling a lemon
   D. Intake of excess sugar and salts
   E. Playing music.

vii. Mr. and Mrs. Mwenda have normal skin but they have four children and only one of them was an albino. What are their possible genotypes?

A. AA x AA
B. aa x aa
C. Aa x Aa
D. AA x Aa
E. Aa x aa

viii. Human body is very sensitive to both internal and external changes that can be explained in different ways. Which of the following is termed as an allergy?

A. Body reactions to antigen that do not bother most people
B. Body reaction against a specific drug
C. Body reaction against a disease-causing micro-organism
D. Accumulation of mucus by inhaling dust particles
E. Formation of rashes caused by eating specific food.

ix. During the biology class session, a biology teacher explained to his students about a theory of origin of life which states that, "The universe has always existed and has no origin, that is it did not have a moment of creation thus life has no origin". Which of the following is the correct name of this theory?

A. The steady state theory
B. Special creation theory
C. Organic evolution theory
D. Cosmozoan theory
E. Spontaneous generation theory.

x. Vascular plants have the specialized tissues for transportation of materials in different parts of the plant. Explain what will happen if phloem tissue is destroyed in green plant?

A. Absorption of water in plant body will stop
B. Absorption of mineral salt and water will stop
C. Transport of manufactured water and food will stop
D. Transport of manufactured food will stop
E. Transport of oxygen in plant body will stop

2. Match the responses in List B with the word/phrases in List A by writing the letter of the correct responses beside the item number

List A	List B
i. Brings about contraction of the uterus at parturition (birth). ii. Causes expulsion of milk from mammary gland. iii. Promotes conversion excess glucose in blood into glycogen. iv. Prepare the body for emergency v. Stimulate kidneys to absorb water. vi. Stimulates development of sexual secondary characteristics in female	A. Anti-diuretic hormone (ADH) B. Oestrogen C. Insulin hormone D. Glucagon hormone E. Oxytocin hormone F. Progesterone G. Adrenal cortical hormones. H. Adrenaline and Noradrenalin I. Androgen

SECTION B (54MARKS)
Answer all questions from this section

3. In the Biology laboratory there different apparatus and equipment used for conducting experiments. Name and draw the apparatus used for;
    i. Holding and heating small sample of liquid.
    ii. Catching small insects.
    iii. Magnify specimens.
    iv. Determining temperature of liquids.
    v. Pick up small organism such as insects without harming them.
    vi. Washing test tube

4. a) Explain in one sentence why the following features occur to these organisms.
    i. Streamlined body of birds.
    ii. Ciliated trachea of human.
    iii. Needle-like leaves of plants found in desert.
    iv. Coiled and elongated ileum in humans

(b) Many people especially the youth engage in bad habits of using drugs such as cocaine, heroin, and marijuana.

    i. What do you think are causes of the youth to engage in drug abuse? (give three points)
    ii. Suggest the ways in which we can stop youth from this bad habit. (give two points)

5. One of the features of living organisms is ability to exchange of oxygen and carbon dioxide across a respiratory surface.
    a) Explain factors that affecting the rate of gaseous exchange in human being (give five points)
    b) Explain why the process is important for the survival of living things?

6. In hot weather humans tend to produce more sweat and a smaller volume of darker colored urine. In cold weather humans tend to produce more urine that is paler in colour.
    a) Use your knowledge of how the kidney functions to explain these observations.
    b) Explain four factors that determine the amount and colour of urine.

7. Blood circulation is the movement of blood from the heart to all parts of the body and back to the heart.
    a) Give any four components of blood
    b) Outline five functions of blood.
    c) Explain why the wall of the left ventricle is thicker than that of the right ventricle
    d) Explain the role of;
        i. Valves
        ii. Septum

8. In multicellular organism the characteristics are carried in the nucleus and are considered to be genetic materials as explained in genetic studies.
    a) Give four differences between DNA and RNA
    b) Colour blindness is a sex linked character controlled by a recessive gene located on the X chromosome. Only homozygous females suffer from this condition and it is more common to males. What percentage will the males suffering from colour blindness be if a normal male marries a colour blind female? (Show a net crossing)

SECTION C (30 MARKS)

Answer only two questions from this section

9. The process whereby food substances are broken down to release energy in the absence of oxygen is called anaerobic respiration.
    a) Explain six applications of this process in your daily life.
    b) Give four differences between anaerobic respiration and anaerobic respiration.

10. Cell division result into gamete cells and production of somatic cell. Explain with help of diagrams the phases that occur during division of somatic cells

11. Recently most women are using artificial methods to prevent pregnancies intentionally without considering the negative impacts of these methods. Assume you are the health officer of Makutupora village prepare a report that explain in details about the negative impacts of these methods of family planning (give six strong points)